Execise Sheet 5: SEQUENCES SERIES

January 4, 2026

Infinite Sequences

Question 1

$$a_n = 1 - (0.2)^n$$

Solution

$$a_n = 1 - (0.2)^n,$$

so the sequence is

$$\{0.8,\ 0.96,\ 0.992,\ 0.9984,\ 0.99968, \ldots\}.$$

Question 2

$$a_n = \frac{n+1}{3n-1}$$

Solution

$$a_n = \frac{n+1}{3n-1},$$

so the sequence is

$$\left\{\frac{2}{2}, \frac{3}{5}, \frac{4}{8}, \frac{5}{11}, \frac{6}{14}, \ldots\right\} = \left\{1, \frac{3}{5}, \frac{1}{2}, \frac{5}{11}, \frac{3}{7}, \ldots\right\}.$$

Question 3

$$a_1 = 3, \qquad a_{n+1} = 2a_n - 1$$

Solution

$$a_2 = 2(3) - 1 = 5, \quad a_3 = 2(5) - 1 = 9, \quad a_4 = 2(9) - 1 = 17, \quad a_5 = 2(17) - 1 = 33.$$

The sequence is

$$\{3, 5, 9, 17, 33, \ldots\}.$$

Question 4

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots\right\}$$

Solution

$$a_n = \frac{1}{2^n}.$$

Question 5

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \ldots\right\}$$

Solution

$$a_n = \frac{1}{2n}.$$

Question 6

$$a_n = n(n-1)$$

Solution

$$a_n \to \infty \quad \text{as } n \to \infty,$$

so the sequence diverges.

Question 7

$$a_n = \frac{n+1}{3n-1}$$

Solution

$$a_n = \frac{1+\frac{1}{n}}{3-\frac{1}{n}},$$

so

$$a_n \to \frac{1}{3} \quad \text{as } n \to \infty.$$

Converges.

Question 8

$$a_n = \frac{3+5n^2}{n+n^2}$$

Solution

$$a_n = \frac{(3+5n^2)/n^2}{(n+n^2)/n^2} = \frac{5+\frac{3}{n^2}}{1+\frac{1}{n}},$$

so

$$a_n \to \frac{5+0}{1+0} = 5 \quad \text{as } n \to \infty.$$

Converges.

Question 9

$$a_n = \frac{\sqrt{n}}{1+\sqrt{n}}$$

Solution

$$a_n = \frac{1}{1+\frac{1}{\sqrt{n}}},$$

so

$$a_n \to \frac{1}{0+1} = 1 \quad \text{as } n \to \infty.$$

Converges.

Question 10

$$a_n = \frac{2^n}{3^{n+1}}$$

Solution

$$a_n = \frac{1}{3}\left(\frac{2}{3}\right)^n,$$

so

$$\lim_{n\to\infty} a_n = \frac{1}{3}\lim_{n\to\infty}\left(\frac{2}{3}\right)^n = \frac{1}{3}\cdot 0 = 0,$$

by with $r = \frac{2}{3}$. Converges.

Question 11

$$a_n = \frac{n}{1+\sqrt{n}}$$

Solution The numerator approaches ∞ and the denominator approaches $0+1=1$ as $n \to \infty$, so

$$a_n \to \infty \quad \text{as } n \to \infty,$$

and the sequence diverges.

Question 12

$$a_n = \frac{(-1)^{n-1}n}{n^2+1}$$

Solution

$$0 \leq |a_n| = \frac{n}{n^2+1} \leq \frac{1}{n} \to 0 \quad \text{as } n \to \infty,$$

so $a_n \to 0$ by the Squeeze Theorem and if $|a_n| \to 0$, then $a_n \to 0$ converges .

Question 13

$$a_n = \frac{(-1)^n n^3}{n^3+2n^2+1}$$

Solution

$$|a_n| = \frac{n^3}{n^3+2n^2+1} = \frac{1}{1+\frac{2}{n}+\frac{1}{n^3}} \to 1 \quad \text{as } n \to \infty,$$

but the terms alternate in sign, so the subsequences $\{a_1, a_3, a_5, \ldots\}$ and $\{a_2, a_4, a_6, \ldots\}$ converge to -1 and 1, respectively. The sequence diverges.

Question 14

$$a_n = \cos\left(\frac{n}{2}\right)$$

Solution The sequence diverges since the terms do not approach any particular real number as $n \to \infty$. The terms take on values between -1 and 1.

Question 15

Determine whether the sequence is increasing, decreasing, or not monotonic. Is the sequence bounded?

$$a_n = \frac{1}{5^n}$$

Solution

$$a_n = \frac{1}{5^n}$$

defines a decreasing geometric sequence since

$$a_{n+1} = \frac{1}{5}a_n < a_n \quad \text{for each } n \geq 1.$$

The sequence is bounded since

$$0 < a_n \leq \frac{1}{5} \quad \text{for all } n \geq 1.$$

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Question 16

Determine whether the sequence is increasing, decreasing, or not monotonic. Is the sequence bounded?

$$a_n = \frac{1}{2n+3}$$

Solution

$$a_n = \frac{1}{2n+3}$$

is decreasing since

$$a_{n+1} = \frac{1}{2(n+1)+3} = \frac{1}{2n+5} < \frac{1}{2n+3} = a_n \quad \text{for each } n \geq 1.$$

The sequence is bounded since

$$0 < a_n \leq \frac{1}{5} \quad \text{for all } n \geq 1.$$

Note that $a_1 = \frac{1}{5}$.

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Question 17

Determine whether the sequence is increasing, decreasing, or not monotonic. Is the sequence bounded?

$$a_n = \frac{2n-3}{3n+4}$$

Solution

$$a_n = \frac{2n-3}{3n+4}$$

defines an increasing sequence since for

$$f(x) = \frac{2x-3}{3x+4},$$

we have

$$f'(x) = \frac{(3x+4)(2)-(2x-3)(3)}{(3x+4)^2} = \frac{17}{(3x+4)^2} > 0.$$

The sequence is bounded since

$$a_n > a_1 = -\frac{1}{7} \quad \text{for } n \geq 1,$$

and

$$a_n < \frac{2n}{3n} = \frac{2}{3} \quad \text{for } n \geq 1.$$

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Question 18

Determine whether the sequence is increasing, decreasing, or not monotonic. Is the sequence bounded?

$$a_n = \cos\left(\frac{n\pi}{2}\right)$$

Solution

$$a_n = \cos\left(\frac{n\pi}{2}\right)$$

is not monotonic. The first few terms are

$$0, -1, 0, 1, 0, -1, 0, 1, \ldots$$

In fact, the sequence consists of the terms $0, -1, 0, 1$ repeated over and over again in that order. The sequence is bounded since

$$|a_n| \leq 1 \quad \text{for all } n \geq 1.$$

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Question 19

Show that the sequence defined by

$$a_1 = 2, \qquad a_{n+1} = \frac{1}{3 - a_n}$$

is convergent, and find its limit.

Solution We use induction. Let P_n be the statement that

$$0 < a_n \leq 2.$$

Clearly P_1 is true, since

$$a_2 = \frac{1}{3-2} = 1.$$

Now assume that P_n is true. Then

$$a_{n+1} = \frac{1}{3 - a_n} \leq \frac{1}{3-0} = \frac{1}{3} \leq 2.$$

Also $a_{n+1} > 0$ since $3 - a_n$ is positive. Thus P_{n+1} is true.

To find the limit, we use the fact that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} a_{n+1} = L = \frac{1}{3-L}.$$

Thus

$$L^2 - 3L + 1 = 0,$$

so

$$L = \frac{3 \pm \sqrt{5}}{2}.$$

But $L \leq 2$, so we must have

$$L = \frac{3 - \sqrt{5}}{2}.$$

Series

Question 1

Let

$$a_n = \frac{2n}{3n+1}.$$

(a) Determine whether $\{a_n\}$ is convergent.
(b) Determine whether $\sum_{n=1}^{\infty} a_n$ is convergent.

Solution (a)

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{2n}{3n+1} = \frac{2}{3},$$

so the sequence $\{a_n\}$ is convergent by
(b) Since

$$\lim_{n\to\infty} a_n = \frac{2}{3} \neq 0,$$

the series $\sum_{n=1}^{\infty} a_n$ is divergent by the nth term test for divergence.

Question 2

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$3+2+\frac{4}{3}+\frac{8}{9}+\cdots$$

Solution

$$3+2+\frac{4}{3}+\frac{8}{9}+\cdots$$

is a geometric series with first term $a=3$ and common ratio $r=\frac{2}{3}$. Since

$$|r|=\frac{2}{3}<1,$$

the series converges to

$$\frac{a}{1-r}=\frac{3}{1-2/3}=\frac{3}{1/3}=9.$$

Question 3

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\frac{1}{8}-\frac{1}{4}+\frac{1}{2}-1+\cdots$$

Solution

$$\frac{1}{8}-\frac{1}{4}+\frac{1}{2}-1+\cdots$$

is a geometric series with $r=-2$. Since

$$|r|=2>1,$$

the series diverges.

Question 4

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$-2+\frac{5}{2}-\frac{25}{8}+\frac{125}{32}-\cdots$$

Solution

$$-2+\frac{5}{2}-\frac{25}{8}+\frac{125}{32}-\cdots$$

is a geometric series with $a=-2$ and $r=\frac{5/2}{-2}=-\frac{5}{4}$. Since

$$|r|=\frac{5}{4}>1,$$

the series diverges by (4).

Question 5

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$1+0.4+0.16+0.064+\cdots$$

Solution

$$1+0.4+0.16+0.064+\cdots$$

is a geometric series with ratio 0.4. The series converges to

$$\frac{a}{1-r}=\frac{1}{1-0.4}=\frac{1}{1-2/5}=\frac{5}{3}$$

since

$$|r|=\frac{2}{5}<1.$$

Question 6

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} 5\left(\frac{2}{3}\right)^{n-1}$$

Solution

$$\sum_{n=1}^{\infty} 5\left(\frac{2}{3}\right)^{n-1}$$

is a geometric series with $a = 5$ and $r = \frac{2}{3}$. Since $|r| = \frac{2}{3} < 1$, the series converges to

$$\frac{a}{1-r} = \frac{5}{1-2/3} = \frac{5}{1/3} = 15.$$

Question 7

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{(-6)^{n-1}}{5^{n-1}}$$

Solution

$$\sum_{n=1}^{\infty} \frac{(-6)^{n-1}}{5^{n-1}}$$

is a geometric series with $a = 1$ and $r = -\frac{6}{5}$. The series diverges since

$$|r| = \left|\frac{6}{5}\right| > 1.$$

Question 8

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n}$$

Solution

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n} = \frac{1}{4}\sum_{n=1}^{\infty}\left(-\frac{3}{4}\right)^{n-1}.$$

The latter series is geometric with $a = 1$ and $r = -\frac{3}{4}$. Since $|r| = \frac{3}{4} < 1$, it converges to

$$\frac{1}{1-(-3/4)} = \frac{1}{7/4} = \frac{4}{7}.$$

Thus, the given series converges to

$$\left(\frac{1}{4}\right)\left(\frac{4}{7}\right) = \frac{1}{7}.$$

Question 9

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=0}^{\infty} \frac{1}{(\sqrt{2})^n}$$

Solution

$$\sum_{n=0}^{\infty} \frac{1}{(\sqrt{2})^n}$$

is a geometric series with ratio $r = \frac{1}{\sqrt{2}}$. Since $|r| = \frac{1}{\sqrt{2}} < 1$, the series converges. Its sum is

$$\frac{1}{1-1/\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2}-1} = \frac{\sqrt{2}}{\sqrt{2}-1}\cdot\frac{\sqrt{2}+1}{\sqrt{2}+1} = \sqrt{2}(\sqrt{2}+1) = 2+\sqrt{2}.$$

Question 10

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=0}^{\infty} \frac{\pi^n}{3^{n+1}}$$

Solution

$$\sum_{n=0}^{\infty} \frac{\pi^n}{3^{n+1}} = \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{\pi}{3}\right)^n$$

is a geometric series with ratio $r = \frac{\pi}{3}$. Since $|r| > 1$, the series diverges.

Question 11

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}}$$

Solution

$$\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}} = 3 \sum_{n=1}^{\infty} \left(\frac{e}{3}\right)^n$$

is a geometric series with first term $3(e/3) = e$ and ratio $r = \frac{e}{3}$. Since $|r| < 1$, the series converges. Its sum is

$$\frac{e}{1 - e/3} = \frac{3e}{3 - e}.$$

Question 12

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{n}{n+5}$$

Solution

$$\sum_{n=1}^{\infty} \frac{n}{n+5}$$

diverges since

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{n+5} \neq 0.$$

Question 13

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{3}{n}$$

Solution

$$\sum_{n=1}^{\infty} \frac{3}{n} = 3 \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges since its partial sum is 3 times the corresponding partial sum of the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$, which diverges. In general, constant multiples of divergent series are divergent.

Question 14

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1}$$

Solution Using partial fractions, the partial sums are

$$s_n = \sum_{i=2}^{n} \frac{2}{(i-1)(i+1)} = \sum_{i=2}^{n} \left(\frac{1}{i-1} - \frac{1}{i+1} \right)$$

$$= \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n+1}\right).$$

This sum is a telescoping series and

$$s_n = 1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}.$$

Thus

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1} = \lim_{n\to\infty} \left(1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}\right) = \frac{3}{2}.$$

Question 15

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{(n+1)^2}{n(n+2)}$$

Solution

$$\sum_{n=1}^{\infty} \frac{(n+1)^2}{n(n+2)}$$

diverges by (7), the Test for Divergence, since

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{(n+1)^2}{n(n+2)} = \lim_{n\to\infty} \frac{1 + \frac{2}{n} + \frac{1}{n^2}}{1 + \frac{2}{n}} \neq 0.$$

Question 16

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{k=2}^{\infty} \frac{k^2}{k^2-1}$$

Solution

$$\sum_{k=2}^{\infty} \frac{k^2}{k^2-1}$$

diverges by the Test for Divergence since

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} \frac{k^2}{k^2-1} \neq 0.$$

Question 17

Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{2}{n^2+4n+3}$$

Solution Converges.

$$s_n = \sum_{i=1}^{n} \frac{2}{i^2+4i+3} = \sum_{i=1}^{n} \left(\frac{1}{i+1} - \frac{1}{i+3} \right) \quad \text{(using partial fractions).}$$

The latter sum is

$$\left(\frac{1}{2}-\frac{1}{4}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{4}-\frac{1}{6}\right)+\cdots+\left(\frac{1}{n+1}-\frac{1}{n+3}\right)$$

(telescoping series). Thus,

$$\sum_{n=1}^{\infty} \frac{2}{n^2+4n+3} = \lim_{n\to\infty}\left(\frac{1}{2}+\frac{1}{3}-\frac{1}{n+2}-\frac{1}{n+3}\right) = \frac{1}{2}+\frac{1}{3} = \frac{5}{6}.$$

Question 18

Find the values of x for which the series converges. Find the sum of the series for those values of x.

$$\sum_{n=1}^{\infty} \frac{x^n}{3^n}$$

Solution

$$\sum_{n=1}^{\infty} \frac{x^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{x}{3}\right)^n$$

is a geometric series with $r = \frac{x}{3}$, so the series converges $\Leftrightarrow |r| < 1 \Leftrightarrow \frac{|x|}{3} < 1 \Leftrightarrow |x| < 3$; that is, $-3 < x < 3$. In that case, the sum of the series is

$$\frac{a}{1-r} = \frac{x/3}{1-x/3} = \frac{x/3}{(3-x)/3} = \frac{x}{3-x}.$$

Question 19

Find the values of x for which the series converges. Find the sum of the series for those values of x.

$$\sum_{n=1}^{\infty} (x-4)^n$$

Solution

$$\sum_{n=1}^{\infty} (x-4)^n$$

is a geometric series with $r = x-4$, so the series converges $\Leftrightarrow |r| < 1 \Leftrightarrow |x-4| < 1 \Leftrightarrow 3 < x < 5$. In that case, the sum of the series is

$$\frac{x-4}{1-(x-4)} = \frac{x-4}{5-x}.$$

Question 20

Find the values of x for which the series converges. Find the sum of the series for those values of x.

$$\sum_{n=0}^{\infty} 4^n x^n$$

Solution

$$\sum_{n=0}^{\infty} 4^n x^n = \sum_{n=0}^{\infty} (4x)^n$$

is a geometric series with $r = 4x$, so the series converges $\Leftrightarrow |r| < 1 \Leftrightarrow 4|x| < 1 \Leftrightarrow |x| < \frac{1}{4}$. In that case the sum of the series is

$$\frac{1}{1-4x}.$$

Question 21

Find the values of x for which the series converges. Find the sum of the series for those values of x.

$$\sum_{n=0}^{\infty} \frac{(x+3)^n}{2^n}$$

Solution

$$\sum_{n=0}^{\infty} \frac{(x+3)^n}{2^n}$$

is a geometric series with $r = \frac{x+3}{2}$, so the series converges $\Leftrightarrow |r| < 1 \Leftrightarrow \frac{|x+3|}{2} < 1 \Leftrightarrow |x+3| < 2 \Leftrightarrow -5 < x < -1$. For these values of x, the sum of the series is

$$\frac{1}{1-(x+3)/2} = \frac{2}{2-(x+3)} = \frac{2}{-x-1} = -\frac{2}{x+1}.$$

Question 22

Find the values of x for which the series converges. Find the sum of the series for those values of x.

$$\sum_{n=0}^{\infty} \frac{\cos^n x}{2^n}$$

Solution

$$\sum_{n=0}^{\infty} \frac{\cos^n x}{2^n}$$

is a geometric series with first term 1 and ratio $r = \frac{\cos x}{2}$, so it converges $\Leftrightarrow |r| < 1$.

But

$$|r| = \left|\frac{\cos x}{2}\right| \le \frac{1}{2}$$

for all x. Thus, the series converges for all real values of x and the sum of the series is

$$\frac{1}{1-(\cos x)/2} = \frac{2}{2-\cos x}.$$

SERIES CONVERGENCE TESTS

Integral Test- p-Test

Use the Integral Test to determine whether the series is convergent or divergent.

Question 1

$$\sum_{n=1}^{\infty} \frac{1}{n^4}$$

Solution The function $f(x) = \dfrac{1}{x^4}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^{\infty} \frac{1}{x^4}\,dx = \lim_{t\to\infty} \int_1^t x^{-4}\,dx = \lim_{t\to\infty} \left[\frac{x^{-3}}{-3}\right]_1^t = \lim_{t\to\infty} \left(-\frac{1}{3t^3} + \frac{1}{3}\right) = \frac{1}{3}.$$

Since the improper integral converges, the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^4}$ is convergent by the Integral Test.

Question 2

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{n}}$$

Solution The function $f(x) = x^{-1/4}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^{\infty} x^{-1/4}\,dx = \lim_{t\to\infty} \int_1^t x^{-1/4}\,dx = \lim_{t\to\infty} \left[\frac{4}{3}x^{3/4}\right]_1^t = \infty.$$

Since the improper integral diverges, the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{n}}$ diverges by the Integral Test.

Question 3

$$\sum_{n=1}^{\infty} \frac{1}{3n+1}$$

Solution The function $f(x) = \dfrac{1}{3x+1}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^{\infty} \frac{1}{3x+1}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{3x+1}\,dx = \lim_{b\to\infty} \left[\frac{1}{3}\ln(3x+1)\right]_1^b = \infty.$$

Since the improper integral diverges, the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{3n+1}$ diverges.

Question 4

$$\sum_{n=1}^{\infty} e^{-n}$$

Solution The function $f(x) = e^{-x}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^{\infty} e^{-x}\, dx = \lim_{b\to\infty} \int_1^b e^{-x}\, dx = \lim_{b\to\infty} \left[-e^{-x}\right]_1^b = e^{-1}.$$

Since the improper integral converges, the series $\sum_{n=1}^{\infty} e^{-n}$ converges.

Question 5

$$\sum_{n=1}^{\infty} ne^{-n}$$

Solution The function $f(x) = xe^{-x}$ is continuous and positive on $[1, \infty)$. Moreover,

$$f'(x) = e^{-x} - xe^{-x} = e^{-x}(1-x) < 0 \quad \text{for } x > 1,$$

so f is decreasing on $[1, \infty)$. Thus, the Integral Test applies.

$$\int_1^{\infty} xe^{-x}\, dx = \lim_{b\to\infty} \int_1^b xe^{-x}\, dx = \lim_{b\to\infty} \left[-xe^{-x} - e^{-x}\right]_1^b = \frac{1}{e}.$$

Since the improper integral converges, the series $\sum_{n=1}^{\infty} ne^{-n}$ converges.

Question 6

$$\sum_{n=1}^{\infty} \frac{n+2}{n+1}$$

Solution The function $f(x) = \dfrac{x+2}{x+1} = 1 + \dfrac{1}{x+1}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^{\infty} f(x)\, dx = \lim_{t\to\infty} \int_1^t \left(1 + \frac{1}{x+1}\right) dx = \lim_{t\to\infty} \left[x + \ln(x+1)\right]_1^t = \infty.$$

Hence, the improper integral diverges and the series $\sum_{n=1}^{\infty} \frac{n+2}{n+1}$ diverges.

Note. Since
$$\lim_{n\to\infty} \frac{n+2}{n+1} = 1,$$
the series also diverges by the Test for Divergence.

Question 7

$1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \frac{1}{125} + \cdots$

Solution

$$1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \frac{1}{125} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n^3}.$$

This is a p-series with $p = 3 > 1$, so it converges by p-test.

Determine whether the series is convergent or divergent.

Question 8

$1 + \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \frac{1}{5\sqrt{5}} + \cdots$

Solution

$$1 + \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \frac{1}{5\sqrt{5}} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}.$$

This is a p-series with $p = \frac{3}{2} > 1$, so it converges by p-test.

Question 9

$\sum_{n=1}^{\infty} \frac{5 - 2\sqrt{n}}{n^3}$

Solution

$$\sum_{n=1}^{\infty} \frac{5 - 2\sqrt{n}}{n^3} = \sum_{n=1}^{\infty} \frac{5}{n^3} - 2\sum_{n=1}^{\infty} \frac{1}{n^{5/2}}.$$

Since $\sum \frac{1}{n^3}$ and $\sum \frac{1}{n^{5/2}}$ both converge (by (1) with $p = 3 > 1$ and $p = \frac{5}{2} > 1$), $\sum_{n=1}^{\infty} \frac{5 - 2\sqrt{n}}{n^3}$ converges.

Question 10

$$\sum_{n=3}^{\infty} \frac{5}{n-2}$$

Solution The function $f(x) = \frac{5}{x-2}$ is continuous, positive, and decreasing on $[3, \infty)$, so we can apply the Integral Test.

$$\int_3^{\infty} \frac{5}{x-2}\,dx = \lim_{t\to\infty} \int_3^t \frac{5}{x-2}\,dx = \lim_{t\to\infty} \big[5\ln(x-2)\big]_3^t = \lim_{t\to\infty} 5\ln(t-2) = \infty.$$

Therefore, the series $\displaystyle\sum_{n=3}^{\infty} \frac{5}{n-2}$ diverges.

Question 11

$$\sum_{n=1}^{\infty} \frac{1}{n^2+4}$$

Solution The function $f(x) = \frac{1}{x^2+4}$ is continuous, positive, and decreasing on $[1, \infty)$, so we can apply the Integral Test.

$$\begin{aligned}\int_1^{\infty} \frac{1}{x^2+4}\,dx &= \lim_{t\to\infty} \int_1^t \frac{1}{x^2+4}\,dx = \lim_{t\to\infty} \left[\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)\right]_1^t \\ &= \frac{1}{2}\left[\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right)\right].\end{aligned}$$

Therefore, the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+4}$ converges.

Question 12

$$\sum_{n=1}^{\infty} \frac{3n+2}{n(n+1)}$$

Solution The function

$$f(x) = \frac{3x+2}{x(x+1)} = \frac{2}{x} + \frac{1}{x+1}$$

is continuous, positive, and decreasing on $[1, \infty)$, so we can apply the Integral Test.

$$\begin{aligned}\int_1^{\infty} \frac{3x+2}{x(x+1)}\,dx &= \lim_{t\to\infty} \int_1^t \left(\frac{2}{x} + \frac{1}{x+1}\right) dx \\ &= \lim_{t\to\infty} \big[2\ln x + \ln(x+1)\big]_1^t = \infty.\end{aligned}$$

Thus, the series $\displaystyle\sum_{n=1}^{\infty} \frac{3n+2}{n(n+1)}$ diverges.

Question 13

$$\sum_{n=1}^{\infty} \frac{n}{n^2+1}$$

Solution The function $f(x) = \frac{x}{x^2+1}$ is continuous and positive on $[1, \infty)$, and

$$f'(x) = \frac{1-x^2}{(x^2+1)^2} < 0 \quad \text{for } x > 1,$$

so f is decreasing. Using the Integral Test,

$$\int_1^{\infty} \frac{x}{x^2+1}\, dx = \lim_{t\to\infty} \left[\frac{1}{2}\ln(x^2+1)\right]_1^t = \infty.$$

Thus, the series $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+1}$ diverges.

Question 14

$$\sum_{n=2}^{\infty} \frac{1}{n^2-4n+5}$$

Solution The function $f(x) = \frac{1}{(x-2)^2+1}$ is continuous, positive, and decreasing on $[2, \infty)$, so the Integral Test applies.

$$\int_2^{\infty} \frac{1}{(x-2)^2+1}\, dx = \lim_{t\to\infty} \left[\tan^{-1}(x-2)\right]_2^t = \frac{\pi}{2}.$$

Thus, the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^2-4n+5}$ converges.

Question 15

$$\sum_{n=1}^{\infty} ne^{-n^2}$$

Solution The function $f(x) = xe^{-x^2}$ is continuous and positive on $[1, \infty)$, and

$$f'(x) = e^{-x^2}(1-2x^2) < 0 \quad \text{for } x > 1,$$

so f is decreasing. Applying the Integral Test,

$$\int_1^{\infty} xe^{-x^2}\, dx = \lim_{t\to\infty} \left[-\frac{1}{2}e^{-x^2}\right]_1^t = \frac{1}{2e}.$$

Since the integral converges, the series converges.

Question 16

$$\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$$

Solution The function $f(x) = \frac{\ln x}{x^2}$ is continuous and positive for $x \geq 2$, and

$$f'(x) = \frac{1 - 2\ln x}{x^3} < 0 \quad \text{for } x \geq 2,$$

so f is decreasing. Using the Integral Test,

$$\int_2^{\infty} \frac{\ln x}{x^2}\, dx = \lim_{t\to\infty} \left[-\frac{\ln x}{x} - \frac{1}{x} \right]_2^t = \frac{\ln 2 + 1}{2}.$$

Thus, the series $\displaystyle\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$ converges.

Comparison Tests

Question 17

Use the Comparison Test or the Limit Comparison Test to determine whether the following series converges or diverges.

Solution

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + n + 1}$$

Since

$$\frac{1}{n^2 + n + 1} < \frac{1}{n^2} \quad \text{for all } n \geq 1,$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is a convergent p-series with $p = 2 > 1$, the given series converges by comparison.

Question 18

$$\sum_{n=1}^{\infty} \frac{2}{n^3 + 4}$$

Solution For all $n \geq 1$,

$$\frac{2}{n^3 + 4} < \frac{2}{n^3}.$$

Since

$$\sum_{n=1}^{\infty} \frac{2}{n^3}$$

is a constant multiple of a convergent p-series with $p = 3 > 1$, the given series converges.

Question 19

$$\sum_{n=1}^{\infty} \frac{5}{2+3^n}$$

Solution For all $n \geq 1$,

$$\frac{5}{2+3^n} < \frac{5}{3^n}.$$

Since

$$\sum_{n=1}^{\infty} \frac{5}{3^n}$$

is a convergent geometric series with ratio $\frac{1}{3}$, the given series converges by comparison.

Question 20

$$\sum_{n=2}^{\infty} \frac{1}{n-\sqrt{n}}$$

Solution For all $n \geq 2$,

$$n - \sqrt{n} < n \quad \Rightarrow \quad \frac{1}{n-\sqrt{n}} > \frac{1}{n}.$$

Since

$$\sum_{n=2}^{\infty} \frac{1}{n}$$

diverges, the given series diverges by comparison.

Question 21

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2}$$

Solution For all $n \geq 1$,

$$\frac{n+1}{n^2} = \frac{1}{n} + \frac{1}{n^2} > \frac{1}{n}.$$

Since the harmonic series diverges, the given series diverges by comparison.

Question 22

$$\sum_{n=1}^{\infty} \frac{4+3^n}{2^n}$$

Solution For all $n \geq 1$,

$$\frac{4+3^n}{2^n} > \frac{3^n}{2^n} = \left(\frac{3}{2}\right)^n.$$

Since

$$\sum_{n=1}^{\infty} \left(\frac{3}{2}\right)^n$$

diverges, the given series diverges by comparison.

Question 23

$$\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^2+1}$$

Solution Since

$$0 \leq \cos^2 n \leq 1,$$

we have

$$0 \leq \frac{\cos^2 n}{n^2+1} \leq \frac{1}{n^2}.$$

Because

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, the given series converges by comparison.

Question 24

$$\sum_{n=1}^{\infty} \frac{n^2-1}{3n^4+1}$$

Solution For all $n \geq 1$,

$$\frac{n^2-1}{3n^4+1} < \frac{n^2}{3n^4} = \frac{1}{3n^2}.$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, the given series converges by comparison.

Question 25

$$\sum_{n=2}^{\infty} \frac{n^2+1}{n^3-1}$$

Solution Divide numerator and denominator by n^3:

$$\frac{n^2+1}{n^3-1} = \frac{\frac{1}{n}+\frac{1}{n^3}}{1-\frac{1}{n^3}}.$$

Thus,

$$\lim_{n\to\infty} \frac{n^2+1}{n^3-1} = 0,$$

but

$$\frac{n^2+1}{n^3-1} > \frac{1}{n}$$

for large n. Since the harmonic series diverges, the given series diverges by comparison.

Question 26

$$\sum_{n=1}^{\infty} \frac{1+\sin n}{10^n}$$

Solution Since $|\sin n| \le 1$,

$$0 \le \frac{1+\sin n}{10^n} \le \frac{2}{10^n}.$$

Because

$$\sum_{n=1}^{\infty} \frac{2}{10^n}$$

is a convergent geometric series, the given series converges by comparison.

Question 27

$$\sum_{n=2}^{\infty} \frac{n-1}{n4^n}$$

Solution For all $n \ge 2$,

$$\frac{n-1}{n4^n} < \frac{1}{4^n}.$$

Since

$$\sum_{n=2}^{\infty} \frac{1}{4^n}$$

is a convergent geometric series, the given series converges by comparison.

Question 28

$$\sum_{n=2}^{\infty} \frac{\sqrt{n}}{n-1}$$

Solution For all $n \geq 2$,

$$\frac{\sqrt{n}}{n-1} > \frac{1}{\sqrt{n}}.$$

Since

$$\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}}$$

is a divergent p-series with $p = \frac{1}{2} \leq 1$, the given series diverges.

Question 29

$$\sum_{n=1}^{\infty} \frac{2+(-1)^n}{n\sqrt{n}}$$

Solution For all $n \geq 1$,

$$\left|\frac{2+(-1)^n}{n\sqrt{n}}\right| \leq \frac{3}{n\sqrt{n}}.$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

converges, the given series converges by comparison.

Question 30

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+1}}$$

Solution For all $n \geq 1$,

$$\frac{1}{\sqrt{n^3+1}} < \frac{1}{n^{3/2}}.$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

converges, the given series converges.

Question 31

$$\sum_{n=1}^{\infty} \frac{1}{2n+3}$$

Solution Use the Limit Comparison Test with $b_n = \frac{1}{n}$:

$$\lim_{n\to\infty} \frac{1/(2n+3)}{1/n} = \frac{1}{2}.$$

Since the harmonic series diverges, the given series diverges.

Question 32

$$\sum_{n=1}^{\infty} \frac{2^n}{1+3^n}$$

Solution For all $n \geq 1$,

$$\frac{2^n}{1+3^n} < \left(\frac{2}{3}\right)^n.$$

Since

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$$

is a convergent geometric series, the given series converges.

Question 33

$$\sum_{n=3}^{\infty} \frac{n+2}{(n+1)^3}$$

Solution Use the Limit Comparison Test with $b_n = \frac{1}{n^2}$:

$$\lim_{n\to\infty} \frac{(n+2)/(n+1)^3}{1/n^2} = 1.$$

Since

$$\sum_{n=3}^{\infty} \frac{1}{n^2}$$

converges, the given series converges.

Question 34

$$\sum_{n=1}^{\infty} \frac{5+2n}{(1+n^2)^2}$$

Solution Use the Limit Comparison Test with $b_n = \frac{1}{n^3}$:

$$\lim_{n\to\infty} \frac{(5+2n)/(1+n^2)^2}{1/n^3} = 2.$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

converges, the given series converges.

Question 35

$$\sum_{n=1}^{\infty} \frac{n^2-5n}{n^3+n+1}$$

Solution Use the Limit Comparison Test with $b_n = \frac{1}{n}$:

$$\lim_{n\to\infty} \frac{(n^2-5n)/(n^3+n+1)}{1/n} = 1.$$

Since the harmonic series diverges, the given series diverges.

Ratio-Root Tests

Apply the ratio test to each of the series below. Does it imply that the series converges or diverges, or is the test inconclusive?

Question 36

$$\sum_{n=1}^{\infty} \frac{2n^3}{5^n}$$

Solution Recall that the ratio test is conducted by analyzing the limit

$$\rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|,$$

where a_n are the terms of the series. Based on the value of this limit, we may be able to conclude whether the series converges or diverges.

First, we calculate the ratio:

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{2(n+1)^3/5^{n+1}}{2n^3/5^n} \right| = \frac{2(n+1)^3}{5^{n+1}} \cdot \frac{5^n}{2n^3} = \frac{(n+1)^3}{5n^3}.$$

Now we take the limit:

$$\rho = \lim_{n\to\infty} \frac{(n+1)^3}{5n^3} = \frac{1}{5} < 1.$$

Since the limit is less than 1, the Ratio Test implies that the series converges.

Question 37

Apply the ratio test to the series

$$\sum_{n=1}^{\infty} \frac{4}{n^6}.$$

Solution

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{4/(n+1)^6}{4/n^6}\right| = \frac{n^6}{(n+1)^6}.$$

Taking the limit,

$$\rho = \lim_{n\to\infty} \frac{n^6}{(n+1)^6} = 1.$$

Since $\rho = 1$, the Ratio Test is inconclusive. However, the series converges since it is a p-series with $p > 1$.

Question 38

Apply the ratio test to the series

$$\sum_{n=1}^{\infty} \frac{(-3)^n}{(2n)!}.$$

Solution

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-3)^{n+1}/(2n+2)!}{(-3)^n/(2n)!}\right| = \frac{3(2n)!}{(2n+2)(2n+1)(2n)!} = \frac{3}{(2n+2)(2n+1)}.$$

Taking the limit,

$$\rho = \lim_{n\to\infty} \frac{3}{(2n+2)(2n+1)} = 0 < 1.$$

Thus, the series converges absolutely.

Question 39

Apply the ratio test to the series

$$\sum_{n=1}^{\infty} \frac{(3n)!}{(n!)^3}.$$

Solution

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(3n+3)!}{(n+1)!(n+1)!(n+1)!} \cdot \frac{(n!)^3}{(3n)!} = \frac{(3n+3)(3n+2)(3n+1)}{(n+1)^3}.$$

The numerator is a polynomial of degree 3 with leading coefficient 27, and the denominator is a polynomial of degree 3 with leading coefficient 1. Therefore,

$$\rho = \lim_{n\to\infty} \frac{(3n+3)(3n+2)(3n+1)}{(n+1)^3} = 27 > 1.$$

Hence, the series diverges.

Question 40

Apply the root test to the series

$$\sum_{n=1}^{\infty} \left(\frac{\sqrt{2n^3+34n+5}}{n^3+7}\right)^n.$$

Solution The root test requires computing

$$\rho = \lim_{n\to\infty} |a_n|^{1/n}.$$

$$\rho = \lim_{n\to\infty} \left|\left(\frac{\sqrt{2n^3+34n+5}}{n^3+7}\right)^n\right|^{1/n} = \lim_{n\to\infty} \frac{\sqrt{2n^3+34n+5}}{n^3+7} = 2 > 1.$$

Therefore, the series diverges.

Question 41

Apply the root test to the series

$$\sum_{n=2}^{\infty} \frac{(\ln(n^2))^n}{n^{2n}}.$$

Solution

$$\rho = \lim_{n\to\infty} \left|\frac{(\ln(n^2))^n}{n^{2n}}\right|^{1/n} = \lim_{n\to\infty} \frac{\ln(n^2)}{n^2}.$$

Using L'Hôpital's Rule,

$$\lim_{n\to\infty} \frac{\ln(n^2)}{n^2} = \lim_{n\to\infty} \frac{2/n}{2n} = \lim_{n\to\infty} \frac{1}{n^2} = 0 < 1.$$

Thus, the series converges.

Question 42

Apply the root test to the series

$$\sum_{n=1}^{\infty} \left(\frac{\sqrt{3-2n+10n^2}}{10n^2+n+7}\right)^n.$$

Solution

$$\rho = \lim_{n\to\infty} \left|\left(\frac{\sqrt{3-2n+10n^2}}{10n^2+n+7}\right)^n\right|^{1/n} = \lim_{n\to\infty} \frac{\sqrt{3-2n+10n^2}}{10n^2+n+7} = 1.$$

Since $\rho = 1$, the Root Test is inconclusive.

Question 43

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{5^n}$$

Solution

Use the Ratio Test. $a_n = (-1)^{n-1}\frac{n}{5^n}$, $\quad |a_n| = \frac{n}{5^n}$.

$$\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n+1}{5n} = \lim_{n\to\infty} \frac{1+\frac{1}{n}}{5} = \frac{1}{5}.$$

Since $\rho < 1$, the series converges absolutely.

—

Question 44

$$\sum_{n=1}^{\infty} \frac{3n+2}{5n^3+1}$$

Solution

Use the Ratio Test.

$$\rho = \lim_{n\to\infty} \frac{3(n+1)+2}{5(n+1)^3+1} \cdot \frac{5n^3+1}{3n+2}$$

$$= \lim_{n\to\infty} \frac{15n^4+\cdots}{15n^4+\cdots} = 1.$$

So the Ratio Test is inconclusive. However, by the Limit Comparison Test with

$$\sum_{n=1}^{\infty} \frac{1}{n^2},$$

the series converges.

—

Question 45

$$\sum_{n=1}^{\infty} \frac{2^n}{n}$$

Solution

Use the Ratio Test.

$$\rho = \lim_{n\to\infty} \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = \lim_{n\to\infty} \frac{2n}{n+1} = 2.$$

Since $\rho > 1$, the series diverges.

—

Question 46

$$\sum_{n=0}^{\infty} \frac{1}{10^n}$$

Solution

Use the Root Test.

$$L = \lim_{n\to\infty} \sqrt[n]{\frac{1}{10^n}} = \lim_{n\to\infty} \frac{1}{10} = \frac{1}{10}.$$

Since $L < 1$, the series converges.

—

Question 47

$$\sum_{k=0}^{\infty} \left(\frac{k}{k+10} \right)^k$$

Solution

Use the Root Test.

$$L = \lim_{k \to \infty} \sqrt[k]{a_k} = \lim_{k \to \infty} \frac{k}{k+10} = 1.$$

So the Root Test is inconclusive.

—

Question 48

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

Solution

Use the Ratio Test.

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \left(\frac{n}{n+1} \right)^n = \left(1 + \frac{1}{n} \right)^{-n}.$$

$$\lim_{n \to \infty} \left(1 + \frac{1}{n} \right)^{-n} = e^{-1}.$$

Since $e^{-1} < 1$, the series converges.

—

Question 49

$$\sum_{n=1}^{\infty} a_n, \quad a_0 = 1, \quad a_{n+1} = \frac{a_n}{n}$$

Solution

Use the Ratio Test.

$$\frac{a_{n+1}}{a_n} = \frac{1}{n} \to 0.$$

Therefore, the series converges.

Question 50

Apply the Root Test to determine whether the following series converges or diverges.

$$\sum_{n=1}^{\infty}\left(\frac{3n+1}{5n-2}\right)^n$$

Solution We apply the Root Test.

$$L=\lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{3n+1}{5n-2}\right)^n\right|}=\lim_{n\to\infty}\frac{3n+1}{5n-2}=\frac{3}{5}.$$

Since $L<1$, the series converges by the Root Test.

Alternating Series

Test the series for convergence or divergence.

Question 1

$$\sum_{n=1}^{\infty}\left(-\frac{1}{3}+\frac{2}{4}-\frac{3}{5}+\frac{4}{6}-\frac{5}{7}+\cdots\right)$$

Solution

$$-\frac{1}{3}+\frac{2}{4}-\frac{3}{5}+\frac{4}{6}-\frac{5}{7}+\cdots=\sum_{n=1}^{\infty}(-1)^n\frac{n}{n+2}.$$

Here $a_n=(-1)^n\frac{n}{n+2}$. Since

$$\lim_{n\to\infty}a_n\neq 0$$

(in fact the limit does not exist), the series diverges by the Test for Divergence.

—

Question 2

$$\sum_{n=1}^{\infty}\left(\frac{4}{7}-\frac{4}{8}+\frac{4}{9}-\frac{4}{10}+\frac{4}{11}-\cdots\right)$$

Solution

$$\frac{4}{7}-\frac{4}{8}+\frac{4}{9}-\frac{4}{10}+\frac{4}{11}-\cdots=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{4}{n+6}.$$

Now $b_n = \frac{4}{n+6} > 0$, $\{b_n\}$ is decreasing, and

$$\lim_{n\to\infty} b_n = 0,$$

so the series converges by the Alternating Series Test.

—

Question 3

$$\sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln n}$$

Solution Let $b_n = \frac{1}{\ln n}$. Then $b_n > 0$, $\{b_n\}$ is decreasing, and

$$\lim_{n\to\infty} b_n = 0.$$

Thus the series converges by the Alternating Series Test.

—

Question 4

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$$

Solution Let $b_n = \frac{1}{\sqrt{n}} > 0$. Then $\{b_n\}$ is decreasing and

$$\lim_{n\to\infty} b_n = 0,$$

so the series converges by the Alternating Series Test.

—

Question 5

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{3n-1}$$

Solution Let $b_n = \frac{1}{3n-1} > 0$. Then $\{b_n\}$ is decreasing and

$$\lim_{n\to\infty} b_n = 0,$$

so the series converges by the Alternating Series Test.

—

Question 6

$$\sum_{n=1}^{\infty}(-1)^n\frac{3n-1}{2n+1}$$

Solution

$$\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}(-1)^n\frac{3n-1}{2n+1}=\sum_{n=1}^{\infty}(-1)^n b_n.$$

Now

$$\lim_{n\to\infty}b_n=\lim_{n\to\infty}\frac{3-\frac{1}{n}}{2+\frac{1}{n}}=\frac{3}{2}\neq 0.$$

Since $\lim_{n\to\infty}a_n\neq 0$ (in fact the limit does not exist), the series diverges by the Test for Divergence.

—

Question 7

$$\sum_{n=1}^{\infty}(-1)^n\frac{2n}{4n^2+1}$$

Solution Let

$$b_n=\frac{2n}{4n^2+1}>0.$$

Then $\{b_n\}$ is decreasing and

$$\lim_{n\to\infty}b_n=\lim_{n\to\infty}\frac{2/n}{4+1/n^2}=0.$$

Therefore, the series converges by the Alternating Series Test.

—

Question 8

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{4n^2+1}$$

Solution Let

$$b_n=\frac{1}{4n^2+1}>0.$$

Then $\{b_n\}$ is decreasing and

$$\lim_{n\to\infty}b_n=0,$$

so the series converges by the Alternating Series Test.

—

Question 9

$$\sum_{n=1}^{\infty}(-1)^n\frac{\sqrt{n}}{1+2\sqrt{n}}$$

Solution

$$\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}(-1)^n\frac{\sqrt{n}}{1+2\sqrt{n}}=\sum_{n=1}^{\infty}(-1)^n b_n.$$

Now

$$\lim_{n\to\infty}b_n=\lim_{n\to\infty}\frac{1}{2+1/\sqrt{n}}=\frac{1}{2}\neq 0.$$

Since $\lim_{n\to\infty}a_n\neq 0$, the series diverges by the Test for Divergence.

—

Question 10

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{n^3+4}$$

Solution Let

$$b_n=\frac{n^2}{n^3+4}>0.$$

Then $\{b_n\}$ is decreasing for $n\geq 2$ since

$$\left(\frac{x^2}{x^3+4}\right)'=\frac{(x^3+4)(2x)-x^2(3x^2)}{(x^3+4)^2}=\frac{x(8-x^3)}{(x^3+4)^2}<0\quad(x>2).$$

Also,

$$\lim_{n\to\infty}b_n=\lim_{n\to\infty}\frac{1/n}{1+4/n^3}=0.$$

Thus the series converges by the Alternating Series Test.

Absolute and Conditionally Convergence

Question 1

Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

$$\sum_{n=1}^{\infty}\frac{n^2}{2^n}$$

Solution The series has positive terms. Using the Ratio Test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\frac{(n+1)^2}{2^{n+1}}\cdot\frac{2^n}{n^2}=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^2\cdot\frac{1}{2}=\frac{1}{2}<1.$$

Therefore, the series is absolutely convergent by the Ratio Test.

—

Question 2

$$\sum_{n=0}^{\infty} \frac{(-10)^n}{n!}$$

Solution Using the Ratio Test,

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{|(-10)^{n+1}|}{(n+1)!} \cdot \frac{n!}{|(-10)^n|} = \lim_{n\to\infty} \frac{10}{n+1} = 0 < 1.$$

Thus, the series is absolutely convergent.

—

Question 3

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^n}{n^4}$$

Solution The series diverges by the Test for Divergence since

$$\lim_{n\to\infty} \frac{2^n}{n^4} = \infty,$$

so

$$\lim_{n\to\infty} (-1)^{n-1} \frac{2^n}{n^4}$$

does not exist.

—

Question 4

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt[4]{n}}$$

Solution The series converges by the Alternating Series Test. However,

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{n}}$$

is a divergent p-series with $p = \frac{1}{4} < 1$. Therefore, the given series is conditionally convergent.

—

Question 5

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^4}$$

Solution The series

$$\sum_{n=1}^{\infty} \frac{1}{n^4}$$

is a convergent p-series with $p = 4 > 1$. Hence,

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^4}$$

is absolutely convergent.

—

Question 6

$$\sum_{n=1}^{\infty} (-1)^n \frac{n}{5+n}$$

Solution

$$\lim_{n\to\infty} |a_n| = \lim_{n\to\infty} \frac{n}{5+n} = 1 \neq 0.$$

Thus, the series diverges by the Test for Divergence.

—

Question 7

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+1}$$

Solution The series

$$\sum_{n=1}^{\infty} \frac{n}{n^2+1}$$

diverges by the Limit Comparison Test with the harmonic series. However, the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+1}$$

has positive terms, is decreasing, and

$$\lim_{n\to\infty} \frac{n}{n^2+1} = 0.$$

Therefore, the series is conditionally convergent by the Alternating Series Test.

—

Question 8

$$\sum_{n=1}^{\infty} \frac{1}{(2n)!}$$

Solution Using the Ratio Test,

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{1/(2n+2)!}{1/(2n)!} = \lim_{n\to\infty} \frac{1}{(2n+2)(2n+1)} = 0 < 1.$$

Hence, the series is absolutely convergent.

—

Question 9

$$\sum_{n=1}^{\infty} e^{-n} n!$$

Solution Using the Ratio Test,

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{e^{-(n+1)}(n+1)!}{e^{-n} n!} = \lim_{n\to\infty} \frac{n+1}{e} = \infty.$$

Thus, the series diverges by the Ratio Test.

—

Question 10

$$\sum_{n=1}^{\infty} (-1)^n \frac{e^{1/n}}{n^3}$$

Solution Since

$$0 \le \frac{e^{1/n}}{n^3} \le \frac{e}{n^3},$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

is a convergent p-series with $p = 3 > 1$, it follows that

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^3}$$

converges. Therefore,

$$\sum_{n=1}^{\infty} (-1)^n \frac{e^{1/n}}{n^3}$$

is absolutely convergent.

Question 11

$$\sum_{n=1}^{\infty} \frac{n^n}{3^{1+3n}}$$

Solution

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \frac{n}{3^3} = \infty,$$

so the series diverges by the Root Test.

Or,

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)^{n+1}}{n^n} \cdot \frac{3^{3n}}{3^{3(n+1)}} = \frac{1}{27} \lim_{n\to\infty} (n+1) = \infty,$$

so the series diverges by the Ratio Test.

Question 12

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$$

Solution Since $\left\{\frac{1}{n \ln n}\right\}$ is decreasing and

$$\lim_{n\to\infty} \frac{1}{n \ln n} = 0,$$

the series converges by the Alternating Series Test.

Since

$$\sum_{n=2}^{\infty} \frac{1}{n \ln n}$$

diverges by the Integral Test, the given series is conditionally convergent.

Question 13

$$\sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+1}\right)^n$$

Solution

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \frac{n^2+1}{2n^2+1} = \frac{1}{2} < 1,$$

so the series is absolutely convergent by the Root Test.

Question 14

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(\arctan n)^n}$$

Solution

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{1}{\arctan n} = \frac{2}{\pi} < 1,$$

so the series is absolutely convergent by the Root Test.

Question 15

$1 - \frac{1\cdot 3}{3!} + \frac{1\cdot 3\cdot 5}{5!} - \frac{1\cdot 3\cdot 5\cdot 7}{7!} + \cdots$

Solution Using the Ratio Test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(2n+1)}{(2n+2)(2n+1)} = 0 < 1,$$

so the series is absolutely convergent.

Question 16

$\frac{2}{5} + \frac{2\cdot 6}{5\cdot 8} + \frac{2\cdot 6\cdot 10}{5\cdot 8\cdot 11} + \cdots$

Solution Using the Ratio Test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{4n+2}{3n+5} = \frac{4}{3} > 1,$$

so the series diverges.

Question 17

$\sum_{n=1}^{\infty}\frac{2\cdot 4\cdot 6\cdots(2n)}{n!}$

Solution Since

$$\frac{2\cdot 4\cdot 6\cdots(2n)}{n!} = 2^n,$$

we have

$$\lim_{n\to\infty} a_n = \infty,$$

so the series diverges by the Test for Divergence.

Question 18

$\sum_{n=1}^{\infty}(-1)^n\frac{2^n n!}{5\cdot 8\cdot 11\cdots(3n+2)}$

Solution Using the Ratio Test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{2(n+1)}{3n+5} = \frac{2}{3} < 1,$$

so the series is absolutely convergent.

Question 19

$a_1 = 2, \quad a_{n+1} = \frac{5n+1}{4n+3}a_n$

Solution By the recursive definition,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{5n+1}{4n+3} = \frac{5}{4} > 1,$$

so the series diverges by the Ratio Test.

Question 20

$a_1 = 1, \quad a_{n+1} = \frac{2+\cos n}{\sqrt{n}} a_n$

Solution

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{|2+\cos n|}{\sqrt{n}} = 0 < 1,$$

so the series converges absolutely by the Ratio Test.

Question 21

For which of the following series is the Ratio Test inconclusive (that is, it fails to give a definite answer)?

1. $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^3}$
2. $\displaystyle\sum_{n=1}^{\infty}\frac{n}{2^n}$
3. $\displaystyle\sum_{n=1}^{\infty}\frac{(-3)^{n-1}}{\sqrt{n}}$
4. $\displaystyle\sum_{n=1}^{\infty}\frac{\sqrt{n}}{1+n^2}$

Solution

1. $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{1/(n+1)^3}{1/n^3} = \lim_{n\to\infty}\frac{n^3}{(n+1)^3} = \lim_{n\to\infty}\frac{1}{(1+1/n)^3} = 1.$$

   Inconclusive.

2. $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+1)/2^{n+1}}{n/2^n} = \lim_{n\to\infty}\frac{n+1}{2n} = \lim_{n\to\infty}\left(\frac{1}{2}+\frac{1}{2n}\right) = \frac{1}{2}.$$

   Conclusive (convergent).

3.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{|(-3)^n|/\sqrt{n+1}}{|(-3)^{n-1}|/\sqrt{n}} = 3\lim_{n\to\infty}\sqrt{\frac{n}{n+1}} = 3\lim_{n\to\infty}\sqrt{\frac{1}{1+1/n}} = 3.$$

Conclusive (divergent).

4.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{\sqrt{n+1}/(1+(n+1)^2)}{\sqrt{n}/(1+n^2)} = \lim_{n\to\infty}\sqrt{\frac{n+1}{n}}\cdot\frac{1+n^2}{1+(n+1)^2}$$

$$= \lim_{n\to\infty}\sqrt{1+\frac{1}{n}}\cdot\frac{1/n^2+1}{1/n^2+(1+1/n)^2} = 1.$$

Inconclusive.

Question 22

For which positive integers k is $\displaystyle\sum_{n=1}^{\infty}\frac{(n!)^2}{(kn)!}$ convergent?

Solution Using the Ratio Test,

$$\lim_{n\to\infty}\frac{(n+1)^2}{k^2} = \begin{cases}\infty, & k=1,\\ \frac{1}{4}, & k=2,\\ 0, & k>2.\end{cases}$$

Thus, the series diverges for $k=1$ and converges for all $k\geq 2$.

Power Series

Find the radius of convergence and interval of convergence of the series.

Question 1

$$\sum_{n=1}^{\infty}\frac{x^n}{\sqrt{n}}$$

Solution If $a_n = \dfrac{x^n}{\sqrt{n}}$, then

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{x^{n+1}}{\sqrt{n+1}}\cdot\frac{\sqrt{n}}{x^n} = \lim_{n\to\infty}|x|\sqrt{\frac{n}{n+1}} = |x|.$$

By the Ratio Test, the series converges when $|x|<1$, so the radius of convergence is $R=1$.

Now we check the endpoints.

When $x = 1$, the series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

diverges because it is a p-series with $p = \frac{1}{2} \leq 1$.

When $x = -1$, the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

converges by the Alternating Series Test.

Thus, the interval of convergence is $[-1, 1)$.

Question 2

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n+1}$$

Solution If $a_n = \dfrac{(-1)^n x^n}{n+1}$, then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{|x|^{n+1}}{n+2} \cdot \frac{n+1}{|x|^n} = |x|.$$

By the Ratio Test, the series converges when $|x| < 1$, so $R = 1$.

When $x = -1$, the series diverges because it is the harmonic series. When $x = 1$, the series becomes the alternating harmonic series, which converges by the Alternating Series Test.

Thus, the interval of convergence is $(-1, 1]$.

Question 3

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n^3}$$

Solution If $a_n = \dfrac{(-1)^{n-1} x^n}{n^3}$, then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)^3} \cdot \frac{n^3}{|x|^n} = |x|.$$

By the Ratio Test, the series converges when $|x| < 1$, so $R = 1$.

Now we check the endpoints.

When $x = 1$, the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3}$$

converges by the Alternating Series Test.

When $x = -1$, the series becomes

$$\sum_{n=1}^{\infty} \frac{1}{n^3},$$

which converges because it is a p-series with $p = 3 > 1$.

Thus, the interval of convergence is $[-1, 1]$.

Question 4

$$\sum_{n=1}^{\infty} \sqrt{n}\, x^n$$

Solution If $a_n = \sqrt{n}\, x^n$, then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{\sqrt{n+1}\, |x|^{n+1}}{\sqrt{n}\, |x|^n} = \lim_{n\to\infty} |x| \sqrt{\frac{n+1}{n}} = |x|.$$

By the Ratio Test, the series converges when $|x| < 1$, so $R = 1$.

When $|x| = 1$,

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \sqrt{n} = \infty,$$

so the series diverges by the Test for Divergence.

Thus, the interval of convergence is $(-1, 1)$.

Question 5

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Solution If $a_n = \dfrac{x^n}{n!}$, then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{|x|^n} = \lim_{n\to\infty} \frac{|x|}{n+1} = 0 < 1.$$

By the Ratio Test, the series converges for all real x.

Thus, $R = \infty$ and the interval of convergence is $(-\infty, \infty)$.

Question 6

$$\sum_{n=1}^{\infty} n^n x^n$$

Solution Here the Root Test is easier.

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{n^n|x|^n} = \lim_{n\to\infty} n|x| = \infty \quad \text{if } x \neq 0.$$

Thus, the series diverges for all $x \neq 0$.

When $x = 0$, the series converges trivially.

Hence, $R = 0$ and the interval of convergence is $\{0\}$.

Question 7

$$\sum_{n=1}^{\infty} (-1)^n 4^n x^n$$

Solution

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)4^{n+1}|x|^{n+1}}{n4^n|x|^n} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right) 4|x| = 4|x|.$$

$$\text{Now } 4|x| < 1 \iff |x| < \frac{1}{4},$$

$$\text{so by the Ratio Test, } R = \frac{1}{4}.$$

When $x = \frac{1}{4}$, we get the divergent series

$$\sum_{n=1}^{\infty} (-1)^n n,$$

and when $x = -\frac{1}{4}$, we get the divergent series

$$\sum_{n=1}^{\infty} n.$$

$$\text{Thus, } I = \left(-\frac{1}{4}, \frac{1}{4}\right).$$

Question 8

$$\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$$

Solution If

$$a_n = \frac{x^n}{n3^n},$$

then

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n}{n+1}\frac{|x|}{3} = \frac{|x|}{3}.$$

By the Ratio Test, the series converges when $|x| < 3$. When $x = 3$, the series becomes $\sum \frac{1}{n}$, which diverges. When $x = -3$, the series becomes $\sum \frac{(-1)^n}{n}$, which converges. Thus, the interval of convergence is $[-3, 3)$.

Question 9

$$\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}}{1 + 2\sqrt{n}} x^n$$

Solution If

$$a_n = (-1)^n \frac{\sqrt{n}}{1 + 2\sqrt{n}} x^n,$$

then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{\sqrt{n+1}}{\sqrt{n}} \frac{1 + 2\sqrt{n}}{1 + 2\sqrt{n+1}} |x| = |x|.$$

By the Ratio Test, the series converges when $|x| < 1$. When $x = \pm 1$, the terms do not approach zero. Thus, the interval of convergence is $(-1, 1)$.

Question 10

$$\sum_{n=1}^{\infty} \frac{(-2)^n x^n}{\sqrt[4]{n}}$$

Solution If

$$a_n = \frac{(-2)^n x^n}{\sqrt[4]{n}},$$

then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} 2|x| \left(\frac{n}{n+1} \right)^{1/4} = 2|x|.$$

By the Ratio Test, the series converges when $|x| < \frac{1}{2}$. When $x = \pm \frac{1}{2}$, the series becomes $\sum \frac{(-1)^n}{\sqrt[4]{n}}$, which diverges. Thus, the interval of convergence is $\left(-\frac{1}{2}, \frac{1}{2}\right)$.

Question 11

$$\sum_{n=1}^{\infty} \frac{x^n}{5^n n^5}$$

Solution If

$$a_n = \frac{x^n}{5^n n^5},$$

then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|}{5}.$$

By the Ratio Test, the series converges when $|x| < 5$. When $x = \pm 5$, the series becomes $\sum \frac{(\pm 1)^n}{n^5}$, which converges. Thus, the interval of convergence is $[-5, 5]$.

Question 12

$$\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n}$$

Solution If

$$a_n = (-1)^n \frac{x^n}{4^n \ln n},$$

then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|}{4}.$$

By the Ratio Test, the series converges when $|x| < 4$. When $x = 4$, the series becomes $\sum \frac{(-1)^n}{\ln n}$, which diverges. When $x = -4$, the series becomes $\sum \frac{1}{\ln n}$, which diverges. Thus, the interval of convergence is $(-4, 4)$.

Question 13

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Solution If

$$a_n = (-1)^n \frac{x^{2n}}{(2n)!},$$

then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{x^2}{(2n+2)(2n+1)} = 0.$$

By the Ratio Test, the series converges for all real x. Thus, the interval of convergence is $(-\infty, \infty)$.

Question 14

$$\sum_{n=0}^{\infty} \sqrt{n}(x-1)^n$$

Solution If

$$a_n = \sqrt{n}(x-1)^n,$$

then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \sqrt{\frac{n+1}{n}}|x-1| = |x-1|.$$

By the Ratio Test, the series converges when $|x-1| < 1$. When $x = 0$ or $x = 2$, the series diverges. Thus, the interval of convergence is $(0, 2)$.

Question 15

$$\sum_{n=0}^{\infty} n^3(x-5)^n$$

Solution If

$$a_n = n^3(x-5)^n,$$

then

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left(\frac{n+1}{n}\right)^3 |x-5| = |x-5|.$$

By the Ratio Test, the series converges when $|x-5| < 1$. When $x = 4$ or $x = 6$, the series diverges. Thus, the interval of convergence is $(4, 6)$.

Representations of Functions as Power Series

Find a power series representation for the function and determine the interval of convergence.

Question 1

$f(x) = \dfrac{1}{1+x}$

Solution Our goal is to write the function in the form $\dfrac{1}{1-r}$, and then use Equation (1) to represent the function as a sum of a power series.

$$f(x) = \frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty}(-x)^n = \sum_{n=0}^{\infty}(-1)^n x^n,$$

which converges for $|x| < 1$. Thus, $R = 1$ and $I = (-1, 1)$.

Question 2

$f(x) = \dfrac{3}{1-x^4}$

Solution

$$f(x) = \frac{3}{1-x^4} = 3\left(\frac{1}{1-x^4}\right) = 3\sum_{n=0}^{\infty}(x^4)^n = \sum_{n=0}^{\infty} 3x^{4n}.$$

The series converges when $|x^4| < 1$, that is, $|x| < 1$. Hence, $R = 1$ and $I = (-1, 1)$.

Question 3

$f(x) = \dfrac{1}{1-x^3}$

Solution Replacing x with x^3 in Equation (1), we get

$$f(x) = \frac{1}{1-x^3} = \sum_{n=0}^{\infty}(x^3)^n = \sum_{n=0}^{\infty} x^{3n}.$$

The series converges when $|x^3| < 1$, that is, $|x| < 1$. Thus, $R = 1$ and $I = (-1, 1)$.

Question 4

$f(x) = \dfrac{1}{1+9x^2}$

Solution

$$f(x) = \frac{1}{1+9x^2} = \frac{1}{1-(-9x^2)} = \sum_{n=0}^{\infty}(-9x^2)^n = \sum_{n=0}^{\infty}(-1)^n 9^n x^{2n}.$$

The series converges when $|-9x^2| < 1$, that is, $|x| < \dfrac{1}{3}$. Hence, $I = \left(-\dfrac{1}{3}, \dfrac{1}{3}\right)$.

Question 5

$f(x) = \dfrac{1}{x-5}$

Solution

$$f(x) = \frac{1}{x-5} = -\frac{1}{5} \cdot \frac{1}{1-\frac{x}{5}} = -\frac{1}{5}\sum_{n=0}^{\infty}\left(\frac{x}{5}\right)^n = -\sum_{n=0}^{\infty}\frac{x^n}{5^{n+1}}.$$

The series converges when $\left|\frac{x}{5}\right| < 1$, that is, $|x| < 5$. Thus, $I = (-5, 5)$.

Question 6

$f(x) = \dfrac{x}{4x+1}$

Solution

$$f(x) = \frac{x}{4x+1} = x \cdot \frac{1}{1-(-4x)} = x\sum_{n=0}^{\infty}(-4x)^n = \sum_{n=0}^{\infty}(-1)^n 4^n x^{n+1}.$$

The series converges when $|-4x| < 1$, that is, $|x| < \frac{1}{4}$. Hence, $I = \left(-\frac{1}{4}, \frac{1}{4}\right)$.

Question 7

$f(x) = \dfrac{x}{9+x^2}$

Solution

$$f(x) = \frac{x}{9+x^2} = \frac{x}{9}\cdot\frac{1}{1+\left(\frac{x}{3}\right)^2} = \frac{x}{9}\sum_{n=0}^{\infty}\left(-\frac{x^2}{9}\right)^n = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{9^{n+1}}.$$

The geometric series converges when

$$\left|\left(\frac{x}{3}\right)^2\right| < 1 \quad\Longleftrightarrow\quad |x| < 3.$$

Thus, $R = 3$ and $I = (-3, 3)$.

Question 8

$f(x) = \dfrac{x^2}{a^3 - x^3}$

Solution

$$f(x) = \frac{x^2}{a^3-x^3} = \frac{x^2}{a^3}\cdot\frac{1}{1-\frac{x^3}{a^3}} = \frac{x^2}{a^3}\sum_{n=0}^{\infty}\left(\frac{x^3}{a^3}\right)^n = \sum_{n=0}^{\infty}\frac{x^{3n+2}}{a^{3n+3}}.$$

The series converges when $\left|\frac{x^3}{a^3}\right| < 1$, that is, $|x| < |a|$. Hence, $R = |a|$ and $I = (-|a|, |a|)$.

Express the function as the sum of a power series by first using partial fractions. Find the interval of convergence.

Question 9

$f(x) = \dfrac{3}{x^2+x-2}$

Solution

$$\frac{3}{x^2+x-2} = \frac{3}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}.$$

Solving gives $A = -1$ and $B = 1$. Hence,

$$\frac{3}{x^2+x-2} = \frac{1}{x-1} - \frac{1}{x+2}.$$

Writing each term as a geometric series,

$$\frac{1}{x-1} = -\frac{1}{1-x} = -\sum_{n=0}^{\infty}x^n, \qquad \frac{1}{x+2} = \frac{1}{2}\cdot\frac{1}{1+\frac{x}{2}} = \sum_{n=0}^{\infty}(-1)^n\frac{x^n}{2^{n+1}}.$$

Thus, the sum converges for $x \in (-1, 1)$.

Question 10

$f(x) = \dfrac{7x-1}{3x^2+2x-1}$

Solution

$$\frac{7x-1}{3x^2+2x-1} = \frac{7x-1}{(3x-1)(x+1)} = \frac{1}{3x-1} + \frac{2}{x+1}.$$

We write

$$\frac{1}{3x-1} = -\frac{1}{1-3x} = -\sum_{n=0}^{\infty}(3x)^n, \qquad \frac{2}{x+1} = 2\cdot\frac{1}{1-(-x)} = 2\sum_{n=0}^{\infty}(-x)^n.$$

The series $\sum(-x)^n$ converges for $|x|<1$, and $\sum(3x)^n$ converges for $|x| < \frac{1}{3}$. Hence, their sum converges for

$$x \in \left(-\frac{1}{3}, \frac{1}{3}\right).$$

Find a power series representation for the function and determine the radius of convergence

Question 11

$f(x) = \ln(5-x)$

Solution

$$f(x) = \ln(5-x) = -\int \frac{dx}{5-x} = \frac{1}{5}\int \frac{dx}{1-\frac{x}{5}}$$

$$= \frac{1}{5}\int \sum_{n=0}^{\infty}\left(\frac{x}{5}\right)^n dx = C - \frac{1}{5}\sum_{n=0}^{\infty}\frac{x^{n+1}}{5^n(n+1)} = C - \sum_{n=1}^{\infty}\frac{x^n}{n5^n}$$

Putting $x=0$, we get $C=\ln 5$. The series converges for $\left|\frac{x}{5}\right| < 1 \iff |x| < 5$, so $R=5$.

—

Question 12

$f(x) = \dfrac{x^2}{(1-2x)^2}$

Solution We know that

$$\frac{1}{1-2x} = \sum_{n=0}^{\infty}(2x)^n.$$

Differentiating, we get
$$\frac{2}{(1-2x)^2} = \sum_{n=1}^{\infty} n2^n x^{n-1} = \sum_{n=0}^{\infty}(n+1)2^{n+1}x^n.$$
Thus,
$$f(x) = \frac{x^2}{(1-2x)^2} = \frac{x^2}{2}\sum_{n=0}^{\infty}(n+1)2^{n+1}x^n = \sum_{n=0}^{\infty}(n+1)2^n x^{n+2}.$$
The radius of convergence is $R = \frac{1}{2}$.
—

Question 13

$f(x) = \dfrac{x^3}{(x-2)^2}$

Solution
$$\frac{1}{2-x} = \frac{1}{2(1-\frac{x}{2})} = \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty}\frac{1}{2^{n+1}}x^n \quad \text{for } |x| < 2.$$
Differentiating,
$$\frac{1}{(x-2)^2} = \frac{d}{dx}\left(\frac{1}{2-x}\right) = \sum_{n=1}^{\infty}\frac{n}{2^{n+1}}x^{n-1} = \sum_{n=0}^{\infty}\frac{n+1}{2^{n+2}}x^n.$$
Thus,
$$f(x) = x^3\sum_{n=0}^{\infty}\frac{n+1}{2^{n+2}}x^n = \sum_{n=0}^{\infty}\frac{n+1}{2^{n+2}}x^{n+3}.$$
Hence, $R = 2$ and $I = (-2, 2)$.
—

Question 14

$f(x) = \arctan\left(\frac{x}{3}\right)$

Solution Using
$$\arctan x = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1}.$$
Thus,
$$f(x) = \arctan\left(\frac{x}{3}\right) = \sum_{n=0}^{\infty}(-1)^n\frac{(x/3)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{3^{2n+1}(2n+1)}.$$
The series converges for $\left|\frac{x}{3}\right| < 1 \iff |x| < 3$, so $R = 3$.

Evaluate the indefinite integral as a power series. What is the radius of convergence?

Question 15

$$\int \frac{t}{1-t^8}\,dt$$

Solution

$$\frac{t}{1-t^8} = \frac{t}{1-(t^8)} = t\sum_{n=0}^{\infty} t^{8n} = \sum_{n=0}^{\infty} t^{8n+1}.$$

Thus,

$$\int \frac{t}{1-t^8}\,dt = \int \sum_{n=0}^{\infty} t^{8n+1}\,dt = C + \sum_{n=0}^{\infty} \frac{t^{8n+2}}{8n+2}.$$

The series converges for $|t| < 1$, so $R = 1$.

Question 16

$$\int \frac{\ln(1-t)}{t}\,dt$$

Solution Using,

$$\ln(1-t) = -\sum_{n=1}^{\infty} \frac{t^n}{n}, \quad |t| < 1.$$

Thus,

$$\frac{\ln(1-t)}{t} = -\sum_{n=1}^{\infty} \frac{t^{n-1}}{n}.$$

Integrating term-by-term,

$$\int \frac{\ln(1-t)}{t}\,dt = C - \sum_{n=1}^{\infty} \frac{t^n}{n^2}.$$

Hence $R = 1$.

Question 17

$$\int \frac{x - \tan^{-1} x}{x^3}\,dx$$

Solution Using,

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad |x| < 1.$$

Thus,

$$x - \tan^{-1} x = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n+1}}{2n+1}.$$

Dividing by x^3,

$$\frac{x - \tan^{-1} x}{x^3} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-2}}{2n+1}.$$

Integrating,

$$\int \frac{x - \tan^{-1} x}{x^3}\, dx = C + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n+1)(2n-1)}.$$

The radius of convergence is $R = 1$.

Question 18

$$\int \tan^{-1}(x^2)\, dx$$

Solution Using,

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad |x| < 1.$$

Substituting x^2 for x,

$$\tan^{-1}(x^2) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+2}}{2n+1}.$$

Integrating,

$$\int \tan^{-1}(x^2)\, dx = C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+3}}{(2n+1)(4n+3)}.$$

The radius of convergence is $R = 1$.

Question 19

$$\int_0^{0.2} \frac{1}{1+x^5}\, dx$$

Solution

$$\frac{1}{1+x^5} = \sum_{n=0}^{\infty} (-1)^n x^{5n}.$$

Thus,

$$\int_0^{0.2} \frac{1}{1+x^5}\, dx = \sum_{n=0}^{\infty} (-1)^n \frac{(0.2)^{5n+1}}{5n+1}.$$

The series is alternating. Using the first two terms,

$$\int_0^{0.2} \frac{1}{1+x^5}\, dx \approx 0.2 - \frac{(0.2)^6}{6} = 0.199989.$$

Question 20

$$\int_0^{0.4} \ln(1+x^4)\,dx$$

Solution Using,

$$\ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}.$$

Replacing x by $-x^4$,

$$\ln(1+x^4) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{4n}}{n}.$$

Integrating,

$$\int_0^{0.4} \ln(1+x^4)\,dx = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(0.4)^{4n+1}}{n(4n+1)}.$$

Using the first three terms,

$$\int_0^{0.4} \ln(1+x^4)\,dx \approx 0.020034.$$

Question 21

$$\int_0^{1/3} x^2 \tan^{-1}(x^4)\,dx$$

Solution Using ,

$$\tan^{-1}(x^4) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{8n+4}}{2n+1}.$$

Thus,

$$x^2 \tan^{-1}(x^4) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{8n+6}}{2n+1}.$$

Integrating,

$$\int_0^{1/3} x^2 \tan^{-1}(x^4)\,dx = \sum_{n=0}^{\infty} (-1)^n \frac{(1/3)^{8n+7}}{(2n+1)(8n+7)}.$$

Using one term,

$$\int_0^{1/3} x^2 \tan^{-1}(x^4)\,dx \approx 0.000065.$$

Question 22

$\int_0^{0.5} \frac{dx}{1+x^6}$

Solution

$$\frac{1}{1+x^6} = \sum_{n=0}^{\infty} (-1)^n x^{6n}.$$

Thus,

$$\int_0^{0.5} \frac{dx}{1+x^6} = \sum_{n=0}^{\infty} (-1)^n \frac{(0.5)^{6n+1}}{6n+1}.$$

Using one term,

$$\int_0^{0.5} \frac{dx}{1+x^6} \approx 0.5.$$

Taylor Maclaurin Series

Find the Maclaurin series for the function $f(x)$ using the definition of a Maclaurin series. Also find the associated radius of convergence.

Question 1

$f(x) = \cos x$

Solution

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\cos x$	1
1	$-\sin x$	0
2	$-\cos x$	-1
3	$\sin x$	0
4	$\cos x$	1

Using the Maclaurin formula,

$$\begin{aligned}\cos x &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}.\end{aligned}$$

If $a_n = \dfrac{(-1)^n x^{2n}}{(2n)!}$, then

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{x^2}{(2n+2)(2n+1)} = 0 < 1.$$

Thus $R = \infty$.

—

Question 2

$f(x) = \sin 2x$

Solution

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\sin 2x$	0
1	$2\cos 2x$	2
2	$-2^2 \sin 2x$	0
3	$-2^3 \cos 2x$	-2^3
4	$2^4 \sin 2x$	0
$\vdots$	$\vdots$	$\vdots$

$$f^{(n)}(0) = 0 \text{ if } n \text{ is even and } f^{(2n+1)}(0) = (-1)^n 2^{2n+1}, \text{ so}$$

$$\sin 2x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{f^{(2n+1)}(0)}{(2n+1)!} x^{2n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n+1}}{(2n+1)!} x^{2n+1}$$

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{2^2 |x|^2}{(2n+3)(2n+2)} = 0 < 1 \text{ for all } x,$$

$$\text{so } R = \infty \quad \text{(Ratio Test).}$$

—

Question 3

$f(x) = (1+x)^{-3}$

Solution

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$(1+x)^{-3}$	1
1	$-3(1+x)^{-4}$	-3
2	$12(1+x)^{-5}$	12
3	$-60(1+x)^{-6}$	-60
4	$360(1+x)^{-7}$	360
$\vdots$	$\vdots$	$\vdots$

$$(1+x)^{-3} = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots$$

$$= 1 - 3x + \frac{4\cdot 3}{2!}x^2 - \frac{5\cdot 4\cdot 3}{3!}x^3 + \frac{6\cdot 5\cdot 4\cdot 3}{4!}x^4 - \cdots$$

$$= 1 - 3x + \frac{4 \cdot 3 \cdot 2}{2 \cdot 2!}x^2 - \frac{5 \cdot 4 \cdot 3 \cdot 2}{2 \cdot 3!}x^3 + \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{2 \cdot 4!}x^4 - \cdots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (n+2)!}{2(n!)} x^n = \sum_{n=0}^{\infty} \frac{(-1)^n (n+2)(n+1)}{2} x^n$$

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(n+3)(n+2)x^{n+1}}{2} \cdot \frac{2}{(n+2)(n+1)x^n} \right|$$

$$= |x| \lim_{n\to\infty} \frac{n+3}{n+1} = |x| < 1 \quad \text{for convergence.}$$

$$\text{So } R = 1 \quad \text{(Ratio Test).}$$

—

Question 4

$f(x) = \ln(1+x)$

Solution

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\ln(1+x)$	0
1	$(1+x)^{-1}$	1
2	$-(1+x)^{-2}$	-1
3	$2(1+x)^{-3}$	2
4	$-6(1+x)^{-4}$	-6
5	$24(1+x)^{-5}$	24
$\vdots$	$\vdots$	$\vdots$

$$\ln(1+x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3$$

$$+ \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots$$

$$= x - \frac{1}{2}x^2 + \frac{2}{6}x^3 - \frac{6}{24}x^4 + \frac{24}{120}x^5 - \cdots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$$

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| = \lim_{n\to\infty} \frac{|x|}{1 + 1/n}$$

$$= |x| < 1 \quad \text{for convergence, so } R = 1.$$

—

Question 5

$f(x) = e^{5x}$

Solution

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	e^{5x}	1
1	$5e^{5x}$	5
2	$25e^{5x}$	25
3	$125e^{5x}$	125
4	$625e^{5x}$	625

Thus,

$$e^{5x} = \sum_{n=0}^{\infty} \frac{5^n}{n!} x^n.$$

Using the Ratio Test,

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = 0,$$

so $R = \infty$.

—

Question 6

$f(x) = xe^x$

Solution

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	xe^x	0
1	$(x+1)e^x$	1
2	$(x+2)e^x$	2
3	$(x+3)e^x$	3

Hence,

$$xe^x = \sum_{n=0}^{\infty} \frac{n}{n!} x^n.$$

Applying the Ratio Test,

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = 0,$$

so $R = \infty$.

Find the Taylor series for centered at the given value of .

Question 7

$f(x) = 1 + x + x^2$, centered at $a = 2$.

Solution

n	$f^{(n)}(x)$	$f^{(n)}(2)$
0	$1 + x + x^2$	7
1	$1 + 2x$	5
2	2	2
3	0	0
4	0	0

$$f(x) = 7 + 5(x-2) + \frac{2}{2!}(x-2)^2$$

Since the derivatives vanish for large n, the radius of convergence is $R = \infty$.

—

Question 8

$f(x) = x^3$, centered at $a = -1$.

Solution

n	$f^{(n)}(x)$	$f^{(n)}(-1)$
0	x^3	-1
1	$3x^2$	3
2	$6x$	-6
3	6	6
4	0	0

$$f(x) = -1 + 3(x+1) - \frac{6}{2!}(x+1)^2 + \frac{6}{3!}(x+1)^3$$

Since the derivatives vanish for large n, the radius of convergence is $R = \infty$.

—

Question 9

$f(x) = e^x$, centered at $a = 3$.

Solution

$$f^{(n)}(x) = e^x \quad \text{for all } n, \qquad f^{(n)}(3) = e^3$$

$$f(x) = \sum_{n=0}^{\infty} \frac{e^3}{n!}(x-3)^n$$

Using the Ratio Test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{|x-3|}{n+1} = 0 < 1$$

for all x, so $R = \infty$.

—

Question 10

$f(x) = \ln x$, centered at $a = 2$.

Solution

n	$f^{(n)}(x)$	$f^{(n)}(2)$
0	$\ln x$	$\ln 2$
1	$\frac{1}{x}$	$\frac{1}{2}$
2	$-\frac{1}{x^2}$	$-\frac{1}{4}$
3	$\frac{2}{x^3}$	$\frac{1}{4}$
4	$-\frac{6}{x^4}$	$-\frac{3}{8}$

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{x^n}, \quad n \geq 1$$

$$\ln x = \ln 2 + \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n\,2^n}(x-2)^n$$

Using the Ratio Test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \frac{|x-2|}{2} < 1$$

so $|x-2| < 2$ and $R = 2$.

—

Question 11

$f(x) = \cos x$, centered at $a = \pi$.

Solution

n	$f^{(n)}(x)$	$f^{(n)}(\pi)$
0	$\cos x$	-1
1	$-\sin x$	0
2	$-\cos x$	1
3	$\sin x$	0
4	$\cos x$	-1

$$\cos x = -1 + \frac{(x-\pi)^2}{2!} - \frac{(x-\pi)^4}{4!} + \frac{(x-\pi)^6}{6!} - \cdots$$

Using the Ratio Test,

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{(x-\pi)^2}{(2n+2)(2n+1)} = 0$$

for all x, so $R = \infty$.

Use a Maclaurin series to obtain the Maclaurin series for the given function

Question 12

$$f(x) = \cos(\pi x)$$

Solution Since

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

we obtain

$$f(x) = \cos(\pi x) = \sum_{n=0}^{\infty} \frac{(-1)^n (\pi x)^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n} x^{2n}}{(2n)!}, \qquad R = \infty.$$

Question 13

$$f(x) = e^{-x/2}$$

Solution Since

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

we have

$$f(x) = \sum_{n=0}^{\infty} \frac{(-x/2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^n, \qquad R = \infty.$$

Question 14

$$f(x) = x \tan^{-1} x$$

Solution Since

$$\tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1},$$

we get

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{2n+1}, \qquad R = 1.$$

Question 15

$$f(x) = \sin(x^4)$$

Solution Since

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

we obtain

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{8n+4}}{(2n+1)!}, \qquad R = \infty.$$

Question 16

$$f(x) = x^2 e^{-x}$$

Solution Using

$$e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!},$$

we have

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+2}}{n!}, \qquad R = \infty.$$

Question 17

$$f(x) = x \cos 2x$$

Solution Since

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

we get

$$\cos(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} x^{2n}}{(2n)!},$$

and hence

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n}}{(2n)!} x^{2n+1}, \qquad R = \infty.$$

Question 18

$$f(x) = \sin^2 x$$

Solution Using

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x),$$

we obtain

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2^{2n-1} x^{2n}}{(2n)!}, \qquad R = \infty.$$

Question 19

$$f(x) = \cos^2 x$$

Solution Using

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x),$$

we obtain

$$f(x) = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n-1} x^{2n}}{(2n)!}, \qquad R = \infty.$$

Question 20

$$f(x) = \frac{\sin x}{x}$$

Solution Since

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

we get

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}, \qquad R = \infty.$$

Question 21

$$f(x) = \frac{x - \sin x}{x^3}$$

Solution Using the Maclaurin series for $\sin x$, we obtain

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+3)!}, \qquad R = \infty.$$

Evaluate the indefinite integral as an infinite series

Question 22

$$\int x \cos(x^3)\, dx$$

Solution We know that

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

Substituting x^3 for x, we obtain

$$\cos(x^3) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{6n}}{(2n)!}.$$

Thus,

$$x\cos(x^3) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n)!}.$$

Integrating term by term,

$$\int x\cos(x^3)\,dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{(6n+2)(2n)!} x^{6n+2}, \qquad R = \infty.$$

Question 23

$$\int \frac{\sin x}{x}\,dx$$

Solution Recall that

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}.$$

Dividing by x gives

$$\frac{\sin x}{x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}.$$

Integrating term by term,

$$\int \frac{\sin x}{x}\,dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)(2n+1)!} x^{2n+1}, \qquad R = \infty.$$

Question 24

$$\int \frac{e^x - 1}{x}\,dx$$

Solution Since

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

we have

$$\frac{e^x - 1}{x} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!}.$$

Integrating term by term,

$$\int \frac{e^x - 1}{x}\, dx = C + \sum_{n=1}^{\infty} \frac{x^n}{n \cdot n!}, \qquad R = \infty.$$